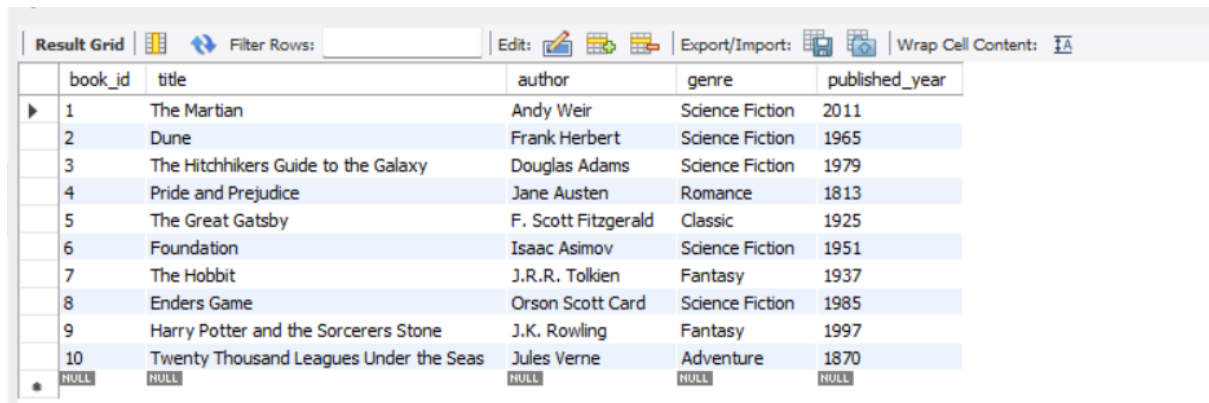


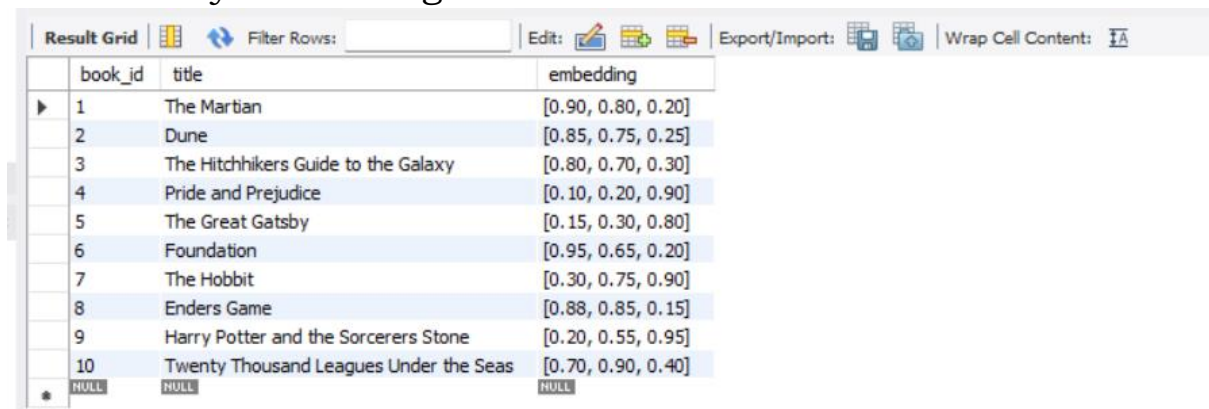
PRACTICAL_4

MySQL Books Table



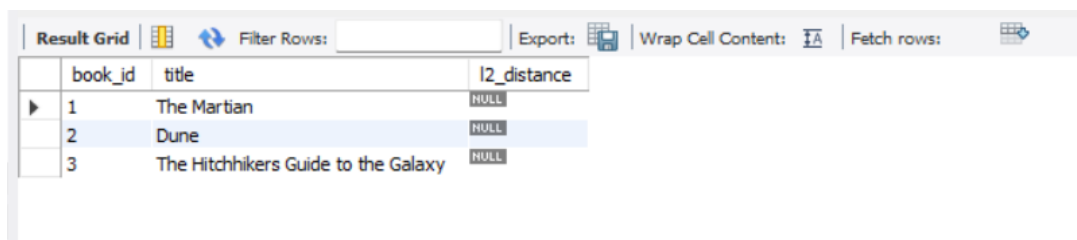
	book_id	title	author	genre	published_year
▶	1	The Martian	Andy Weir	Science Fiction	2011
	2	Dune	Frank Herbert	Science Fiction	1965
	3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
	4	Pride and Prejudice	Jane Austen	Romance	1813
	5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
	6	Foundation	Isaac Asimov	Science Fiction	1951
	7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
	8	Enders Game	Orson Scott Card	Science Fiction	1985
	9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
	10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870
*	NULL	NULL	NULL	NULL	NULL

Add & verify embeddings



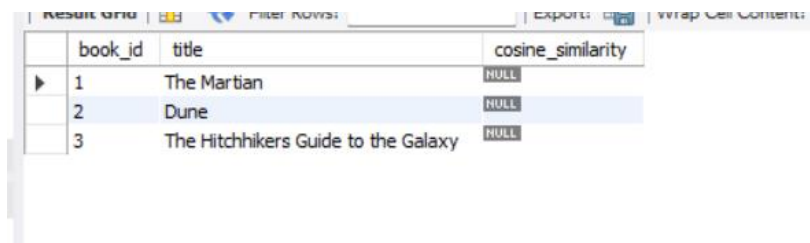
	book_id	title	embedding
▶	1	The Martian	[0.90, 0.80, 0.20]
	2	Dune	[0.85, 0.75, 0.25]
	3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
	4	Pride and Prejudice	[0.10, 0.20, 0.90]
	5	The Great Gatsby	[0.15, 0.30, 0.80]
	6	Foundation	[0.95, 0.65, 0.20]
	7	The Hobbit	[0.30, 0.75, 0.90]
	8	Enders Game	[0.88, 0.85, 0.15]
	9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
	10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]
*	NULL	NULL	NULL

L2 Distance Top 3



	book_id	title	l2_distance
▶	1	The Martian	NULL
	2	Dune	NULL
	3	The Hitchhikers Guide to the Galaxy	NULL

Cosine Similarity



	book_id	title	cosine_similarity
▶	1	The Martian	NULL
	2	Dune	NULL
	3	The Hitchhikers Guide to the Galaxy	NULL

Node.js API

```
Problems Output Terminal Ports
PS C:\Users\uday\OneDrive\Desktop\BookFlow_API> node server.js
code: 'ERR_INVALID_PACKAGE_CONFIG'
}

Node.js v22.19.0
PS C:\Users\uday\OneDrive\Desktop\BookFlow_API> npm install

added 78 packages, and audited 79 packages in 4s

30 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\uday\OneDrive\Desktop\BookFlow_API> node server.js
BookFlow API running at http://localhost:3000
Connected to MySQL
```

API running in browser

```
Pretty-print ☒
{
  "status": "ok",
  "service": "BookFlow Intelligent Catalog API",
  "version": "v1",
  "endpoints": {
    "search": "GET /v1/books/search?q=Stories%20about%20space&limit=3",
    "create": "POST /v1/books"
  }
}
```

Swagger API documentation

```
193 title:
194   type: string
195 author:
196   type: string
197 genre:
198   type: string
199 description:
200   type: string
201 published_year:
202   type: integer
203 embedding:
204   type: array
205   minItems: 3
206   maxItems: 3
207   items:
208     type: number
209 ErrorResponse:
210   type: object
211   properties:
212     error:
213       type: object
214       properties:
```

BookFlow Intelligent Catalog API 1.0.0 OAS 3.0

REST API for semantic book search.

The API searches books using mock three-dimensional vectors stored in MySQL.

Servers

Books Book operations

GET /v1/books/search Search books using semantic similarity

POST /v1/books Create a new book

Schemas

Semantic Search in Swagger

Code	Description	Links
200	Successful search	No links
Media type		
application/json		
Controls Accept header.		
Example Value Schema		
<pre>{ "query": "Stories about space", "count": 3, "results": [{ "book_id": 1, "title": "The Martian", "author": "Andy Weir", "genre": "Science Fiction", "description": "An astronaut struggles to survive alone on Mars.", "published_year": 2011, "similarity": 1 }, { "book_id": 8, "title": "Enders Game", "author": "Orson Scott Card", "genre": "Science Fiction", "description": "A young student is trained for an interstellar conflict.", "published_year": 1985, "similarity": 0.998 }, { "book_id": 2, "title": "Dune", "author": "Frank Herbert", "genre": "Science Fiction", "description": "A young nobleman is trained for an interstellar conflict.", "published_year": 1965, "similarity": 0.998 }] }</pre>		

Missing q → 400 Bad Request

Parameters

Cancel

Name	Description
q * required	Natural language search query
string (query)	q
limit	Maximum number of results
integer (query)	3

Please correct the following validation errors and try again.

- For 'q': Required field is not provided.

Execute

Clear